SUBHANKAR ROY

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SUMMARY

2026 B.Tech Computer Science and Engineering graduate with software engineering experience in Java (Java 17, Spring Boot, Hibernate/JPA) and AI/ML (PyTorch, CUDA, scikit-learn). Built secure, high-performance REST APIs and full-stack features for a government digital services portal and now deliver production web application features end to end. Work across the full software development life cycle in Agile feature teams, with a background in complex analytical and numerical problem solving.

TECHNICAL SKILLS

Languages	Java, Python, TypeScript, SQL, Bash
Java & Backend	Spring Boot, REST API design and integration, FastAPI
AI / ML	PyTorch, scikit-learn
Front End & DB	React, TypeScript, PostgreSQL
DevOps & Cloud	Docker, Git, GitHub Actions (CI/CD), Linux
Testing & Security	JUnit, Postman, OWASP Top 10
Ways of Working	Git-based development and issue tracking, wiki/API documentation, code review

WORK EXPERIENCE

Software Engineer **Oct 2026 – Present**
National Informatics Centre — Agartala, Tripura, India | Full-time

- Build React, Typescript Based Application using Docker for deployment.

Software Engineer **Apr 2026 – Oct 2026**
Hake Technologies — Remote, Tripura, India | Full-time

- Engineer Angular (TypeScript) features for a GIS web application, integrating FastAPI and PostgreSQL REST APIs for seamless data exchange between the client, application server and GeoServer (WMS/WFS).
- Own features across the full life cycle — design, development, enhancement and maintenance — in a permanent Agile team, coordinating API contract changes with backend engineers each sprint.
- Run integration test cycles across front end, backend and GeoServer, diagnosing CORS, layer-loading and coordinate-projection defects before release; build reusable Angular services that remove duplicated request-handling code.

Software Engineering Intern **Jun 2025 – Jul 2025**
National Informatics Centre (NIC), MeitY, Government of India — Tripura | Internship

- Engineered 8+ secure REST endpoints in Java 17 and Spring Boot with Hibernate/JPA and PostgreSQL for Tripura's Professional Tax Portal, covering taxpayer registration and onboarding workflows.
- Implemented JWT authentication and role-based access control; hardened data access with parameterised queries and input sanitisation against SQL injection, in line with OWASP practices.
- Integrated backend APIs with a React/TypeScript front end over a 4-week Agile cycle; resolved cross-origin and data-integrity defects and validated business logic with JUnit and Mockito tests.

AI Computation Research Intern **Jun 2023 – Jul 2023**
Tripura University — Tripura, India | Research Internship

- Benchmarked CNN training across 4 GPU and CPU configurations using PyTorch and CUDA, analysing throughput to recommend an optimal training setup; documented reproducible setup steps and findings for later researchers.

AWS Cloud Trainee **Sep 2021 – Jan 2022**
F13 Technologies — New Delhi, India | Trainee

- Completed structured labs across EC2, S3, RDS, IAM, Lambda and VPC; earned AWS Partner Technical, Business and Cloud Economics accreditations.

PROJECTS

Professional Tax Portal Jun 2025 – Jul 2025
Java 17, Spring Boot, Hibernate/JPA, PostgreSQL, React, TypeScript, JWT, Docker

- Delivered 8+ REST endpoints with JWT authentication and RBAC; containerised the service with Docker and enforced data integrity through database constraints and JUnit/Mockito tests.

Malware Detection on Linux Jul 2025 – Jun 2026
Python, scikit-learn, Bash, strace, procfs

- Built an end-to-end machine learning detection pipeline over system-call traces (`strace`/`procfs`), combining static and dynamic analysis with feature extraction and scikit-learn classification.

Single-Stage RISC Processor — ASIC Flow Feb 2026 – Mar 2026
SystemVerilog, OpenLane, Yosys, OpenROAD

- Implemented the RTL and drove the complete RTL-to-GDSII flow (synthesis, place and route, DRC/LVS) using OpenLane, achieving timing closure at the target frequency.

EDUCATION

B.Tech, Computer Science and Engineering **Jul 2021 – Jun 2026**
NERIST, Arunachal Pradesh, India Coursework: Data Structures and Algorithms, Operating Systems, Networks, DBMS, Computer Architecture

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 Developer Professional (Oct 2025–Oct 2027) | NPTEL: Applied Accelerated Artificial Intelligence (2025); Deep Learning, IIT Ropar | AWS Cloud Practitioner Essentials and AWS Partner Accreditations (2021)